TRACE Implementation Guide: Start to Finish

A step-by-step plan for building, testing and finishing TRACE (Trajectory-based Recovery Assessment with Cause-aware Explainability), from the first decisions to the final report.

The final picture

When TRACE is done, this is what the system looks like:

- A **sock unit**: a sock with a thin pad inside the sole holding 3 pressure sensors (heel, ball of foot, big toe), 1 motion sensor (IMU) on the foot, and an ESP32 in a small box at the ankle.
- A **knee unit**: a knee band with 2 IMUs (one above the knee, one below) and an ESP32 in a small box on the band.
- A **phone app** that connects to both units by Bluetooth, takes the fear questionnaire (TSK-11), records the walk and saves the data.
- A **model on a laptop** that reads each session and gives a rating (Bad, Borderline or Good) and shows why. It also checks whether the cause looks physical or fear-related, and tracks the trend over visits.
- The labels come from a real physio, not from you.

Two terms you will need: **I2C** is the 2-wire connection the ESP32 uses to talk to the IMUs. **BLE** is Bluetooth Low Energy, used to send data to the phone.

Phase 0: Decisions (week 1, no soldering yet)

1. **Fix the scope with your supervisor and physio.** One injury type (ACL is best), one leg measured, one walking test (walk 10 metres, 3 times).
2. **Get the labels in writing.** Ask the physio what Bad, Borderline and Good mean, in a few lines. If possible, get a second physio to rate some sessions too, so you can show the labels are consistent.
3. **Agree the sensor layout.** The proposal says one IMU on the knee. One IMU only tells you the tilt, not how much the knee bends. Two IMUs (above and below the knee) give the knee angle. You have 6 IMUs, so you can afford this. Tell your group and confirm.
4. **Start consent and ethics paperwork now.** Write a consent form in Urdu and English. Use patient IDs like P001, never names. Decide who stores the data. Patients are the slowest part of the project, so start early.

5. **Set up shared tools.** A GitHub repo, a shared drive, and a file naming rule like P001_V1_T1 (patient, visit, trial).
6. **Split the work.** One person on the sock, one on the knee band and firmware, one on the app, one on data and the model. Everyone joins data collection.

Phase 1: Test every part alone on the desk

7. Test both ESP32 boards, as in the first step we discussed. Label them SOCK and KNEE.
8. Test each IMU alone. Wire VCC to 3V3, GND to GND, SDA to GPIO 21 and SCL to GPIO 22. Run an I2C scanner sketch and you should see address 0x68. Tilt the module and watch the numbers change.
9. Test each FSR alone. Each FSR needs a 10 kΩ resistor. Wire 3V3 → FSR → junction → ESP32 pin, and junction → 10 kΩ → GND. Use GPIO 34, 35 and 32. Press it and the reading should rise.
10. Throw away any dead part now. You have spares.

Phase 2: Build the sock unit

11. **Prototype on a breadboard first:** 3 FSRs and 1 IMU on the SOCK ESP32, with all values printed on one line.
12. **Choose the sock.** Buy 2 or 3 identical socks that are thin, stretchy and snug. A loose sock makes the sensors slide and the data noisy.
13. **Make the pad.** Cut a thin foot-shaped pad from felt or foam. Tape each FSR on it, one at the heel, one under the ball of the foot and one under the big toe. Put a small firm disc over each sensing circle so the foot's pressure lands on it. Do not fold the FSR tails sharply.
14. **Put the pad in the sock.** Sew or tape a pocket on the inside of the sole. Mark the exact positions so you can place it the same way every time.
15. **Wires.** Use thin flexible wire, not jumper wires, which break inside a sock. Do not hold a hot iron on an FSR tail, because it melts. Use small connectors or solder very quickly. Cover joints with heat shrink or tape and let the wires run up the side of the sock to the ankle.
16. **Mount the IMU** on top of the foot (the instep) in a pocket, flat and snug, with foam under it. Draw its X/Y/Z directions on it with a marker. Always use the same spot.
17. **Mount the ESP32 box** on a cuff above the ankle bone. Leave some slack in the wires so the ankle can move.

18. **Safety:** the patient should wear flat non-slip shoes over the sock. A sock on a smooth floor is a slip risk.
19. Walk in place for a minute and check that nothing comes loose and the readings are stable.

Phase 3: Build the knee unit

20. Use an elastic knee sleeve or a wide Velcro strap.
21. Put one IMU about 10 to 15 cm above the knee and one about 10 to 15 cm below. Point them the same way and draw arrows on them.
22. Both IMUs share SDA and SCL. Connect the AD0 pin of the second one to 3V3, so it uses address 0x69 and the two do not clash. Run the I2C scanner and you should see 0x68 and 0x69.
23. Put the ESP32 box on the band at the side of the knee. Use short, twisted wires, and keep each IMU wire under about 20 cm.
24. **Calibrate.** The patient stands straight for 5 seconds, which defines 0°. Then bend the knee slowly. The angle should go smoothly up to about 120°.

Phase 4: Firmware (the code on the ESP32)

25. First print everything over USB serial, and only then move to Bluetooth.
26. Read all sensors at 50 readings per second (100 if Bluetooth keeps up). Use a timer, not delay().
27. Each reading holds a sequence number, a timestamp, the 3 FSR values and the IMU accelerometer and gyro values. Send them over BLE in small batches (5 readings per packet). The sequence numbers let you count lost packets later.
28. Name the units TRACE-SOCK and TRACE-KNEE. The phone can send START, STOP and SYNC commands.
29. **Sync the two units.** The phone sends START to both at once, and each unit resets its clock. As a backup, have the patient stamp the heel hard 3 times at the start. The spike appears in both units and lets you align them later.
30. **Buffering.** If Bluetooth drops, save readings to the ESP32's flash (LittleFS) and send them when it reconnects.
31. Send raw values only. Do the filtering and angle maths on the phone or laptop, so you can change it without re-flashing.

Phase 5: Power

32. Use a power bank for all early work.
33. Later, use a protected Li-Po battery, a TP4056 charger module and a boost converter to 5V into the ESP32's 5V pin. Never connect a Li-Po straight to 3V3.
34. Keep the battery in a pouch, not directly against the skin.

Phase 6: Phone app (Flutter)

35. Build these in order: connect to both units, enter patient ID, TSK-11 form, start/stop, live graph, save, upload.
36. TSK-11 is 11 statements with 4 choices each (not a free slider). Total score runs from 11 to 44, higher meaning more fear of movement.
37. Save the raw data as a file per trial, plus a small metadata file (patient ID, visit, date, TSK score).
38. Fix the mockup wording: show "AI Assessment", not "Diagnosis". Show the 3 categories, not a score out of 100. Label the report "AI-generated, for clinician review".

Phase 7: Pilot with friends

39. Test with 3 to 5 healthy volunteers. This only checks that the equipment and your procedure work. Do not use this data for training.
40. Check placement is repeatable, no data is lost, steps are detected, and the whole visit takes a sensible time (about 20 to 30 minutes).
41. Practise the full visit, with a friend playing the physio.

Phase 8: Real data collection (the most important part)

42. Before each visit, charge the units and check the sync.
43. At the visit, take consent, then note: patient ID, age, sex, height, weight, injured leg, injury grade (from the physio or MRI report), weeks since injury or surgery, the weight-bearing instruction, pain level and any walking aid.
44. Fit the sensors and take a photo of the placement.
45. Do the standing calibration. If a patient cannot stand on that leg, do the bench calibration with known weights instead. Do not ask an injured patient to stand on one leg unless the physio says it is fine.

46. The patient fills in TSK-11.
47. The patient walks 10 m, 3 times, with the physio beside them. Start each with the heel stamps.
48. Record video from front, back and side, as your physio asked.
49. **Right after the walks, the physio gives the label** (Bad, Borderline or Good) and, ideally, one line on why. You never label the data yourselves.
50. Remove the sensors, clean them, and back up everything the same day.
51. Try to get as many patients as possible, covering all three labels, not just “Good”. Ask your supervisor what number is acceptable.
52. Ask patients to come back for repeat visits. The trend feature needs at least 3 or 4 visits from a few people.

Phase 9: Cleaning and features

53. Check every file. Align the two units, fill small gaps and remove obvious glitches.
54. Smooth the signals with a low-pass filter. Scale the FSR values by body weight.
55. Detect steps from the heel FSR peaks.
56. Work out per session: cadence, step time, stance time, pressure at heel/ball/toe, knee range of motion, foot roll angle, and mean, spread, minimum and maximum of each. Get stride length and speed from the 10 m distance, the step count and the time. Do not integrate the accelerometer for this, because it drifts.
57. Make one table: one row per session, with patient ID, features, label and TSK score.

Phase 10: The model

58. Start with a baseline: always guess the most common label.
59. Then train an ordinal logistic regression and a random forest. Keep it simple, because the data will be small.
60. **Split by patient, never by session.** Use leave-one-patient-out testing. Otherwise the same person appears in training and testing and the result is fake.
61. Report accuracy, macro-F1, weighted kappa and the confusion matrix. Write down every setting and the split method. The earlier project was criticised for missing exactly this.

Phase 11: Explanation, cause, trend, AI text

62. Add **SHAP**, which shows which features pushed the rating up or down.
63. **Cause layer:** combine the TSK score with knee range of motion. A poor rating with low movement and a high fear score points to “possibly fear-related”. A low fear score points to “more likely physical”. Call it a hint for the physio, not a diagnosis. If your physio can confirm causes on some patients, you can check how often it is right.
64. **Trend:** smooth each patient’s scores over visits with EWMA, and use CUSUM to detect a sudden change or regression.
65. **AI text:** send the rating, top SHAP features, TSK score and trend to an LLM and ask for a plain-language summary. It should flag “consult your doctor” when the trend worsens. It must not diagnose or suggest treatment.

Phase 12: Measure the system

66. Measure battery life, current draw, Bluetooth delay, lost packets (from the sequence numbers), total cost of parts and weight of each unit. The proposal promises these numbers.

Phase 13: Finish

67. Run the full demo end to end. Write the report with the results, the limits (small data, sock placement noise, FSRs measure relative load) and how TRACE differs from the earlier projects.
68. Optional, only if time remains: the piezo energy harvesting your teammates added.

What will probably go wrong

- The sensors shift between visits, so mark positions and photograph every fitting.
- FSR readings are noisy, so calibrate them for each patient.
- Bluetooth drops, which is why we added the buffering.
- Patients are hard to find, so start now.
- Most labels may turn out to be “Good”, so look for the other two early.